



coreon

Knowledge meets language.

Benefits of Collaborative AI: Taxonomization of IATE, the EU Terminology

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The
Knowledge
Graph
Conference

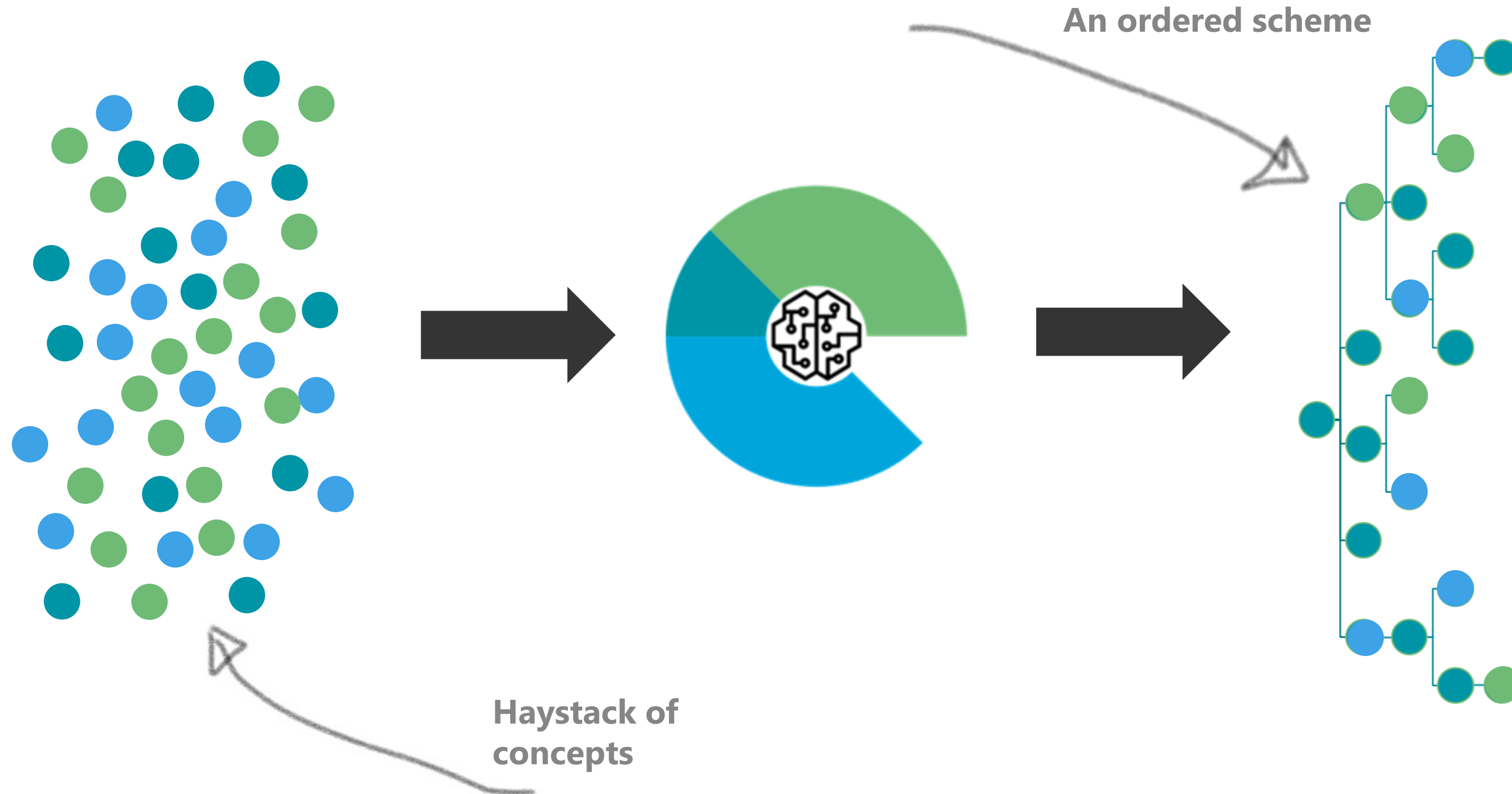
Coreon: Backbone for the Data



- ⌘ convert data into knowledge
- ⌘ incorporate **terminologies**
taxonomies
ontologies
thesauri
vocabularies
into one **Knowledge Graph**
- ⌘ concept-oriented and language-agnostic data model
- ⌘ intuitive, lightweight **UI**



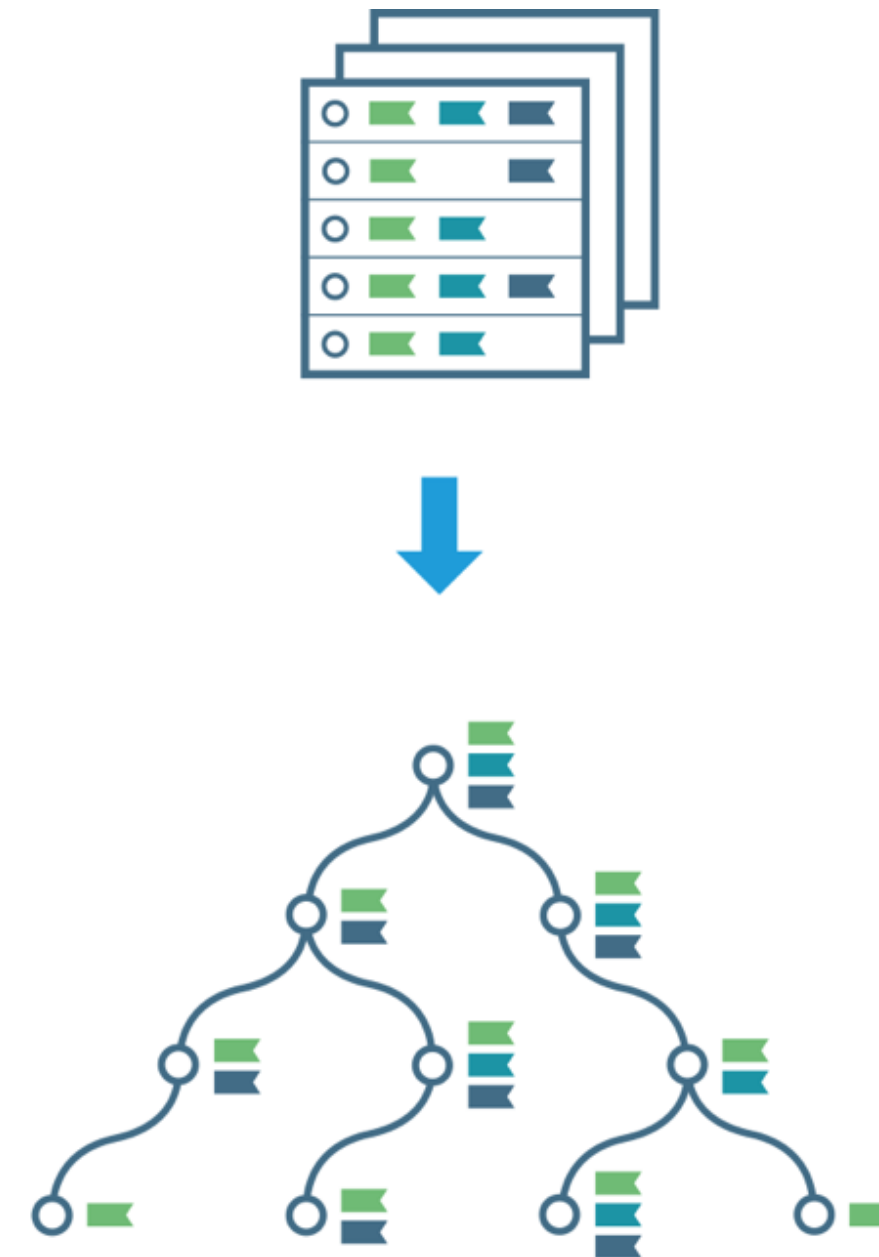
Agenda: Taxonomization in a Nutshell:



- ⌘ Perks of structured data and IATE as a resource
- ⌘ Manual Taxonomization
- ⌘ Automatic Taxonomization, powered by Machine Learning
- ⌘ Benefits of Collaborative-AI Approach
- ⌘ Industry use case: document classification

Perks of Structured Data

- ⌘ A Powerful Resource for AI/ML projects
- ⌘ Cross-lingual Data Analysis
- ⌘ Enterprise Search
- ⌘ Actionable intelligence
- ⌘ Cross-border Interoperability



Interactive Terminology for Europe, IATE

- ⌘ Introduced in 2004, used by most EU Institutions, covers all EU domains
- ⌘ Recent focus on healthcare, financial crisis, environment, fisheries, and migration
- ⌘ EuroVoc for domain classification system

- ⌘ Number of concepts: **961 116**
- ⌘ Number of terms: 7 992 325
- ⌘ New terms last week: 1 646

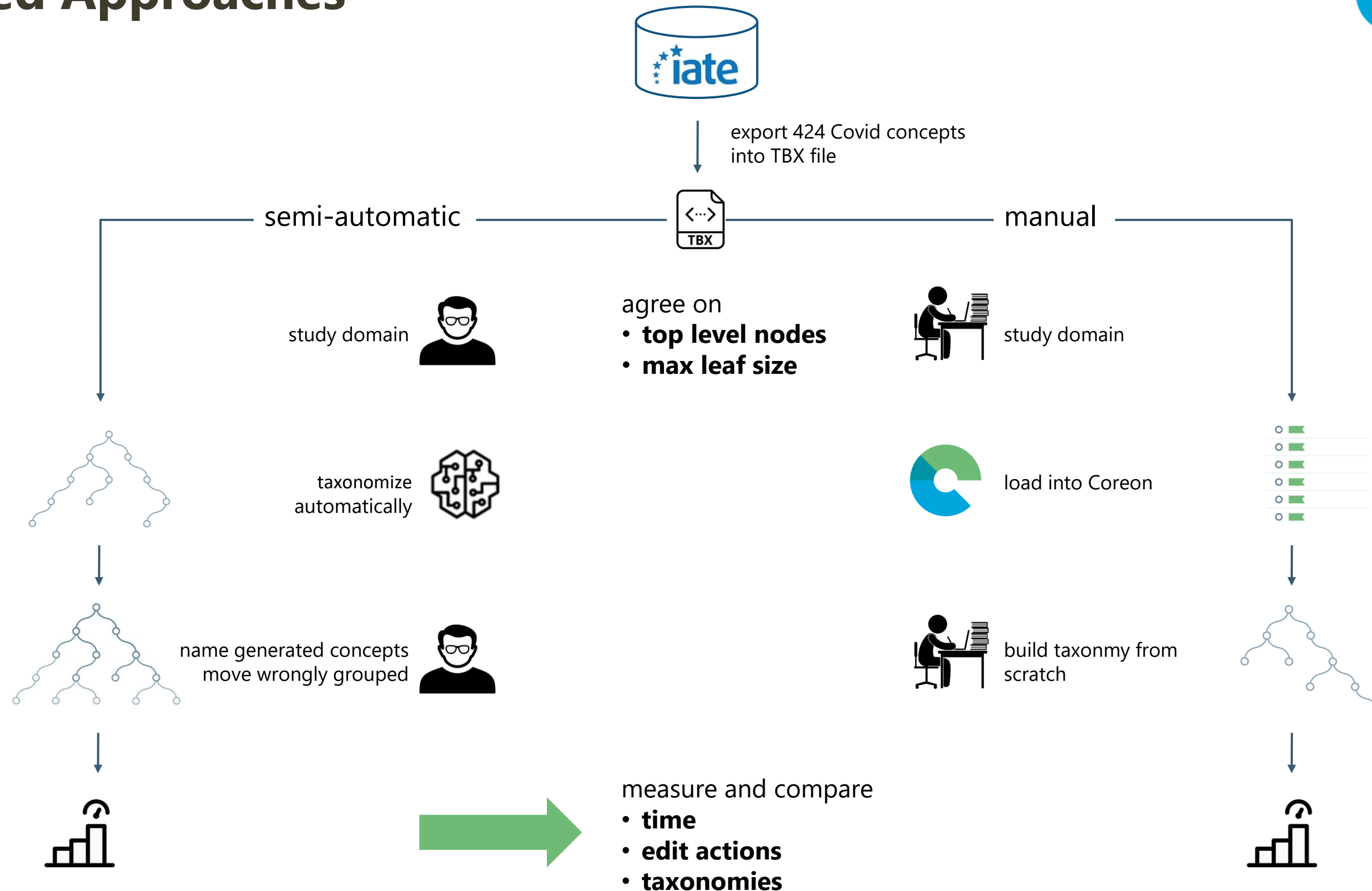


TODO: draft a deeply-structured taxonomy, **_fast_** 

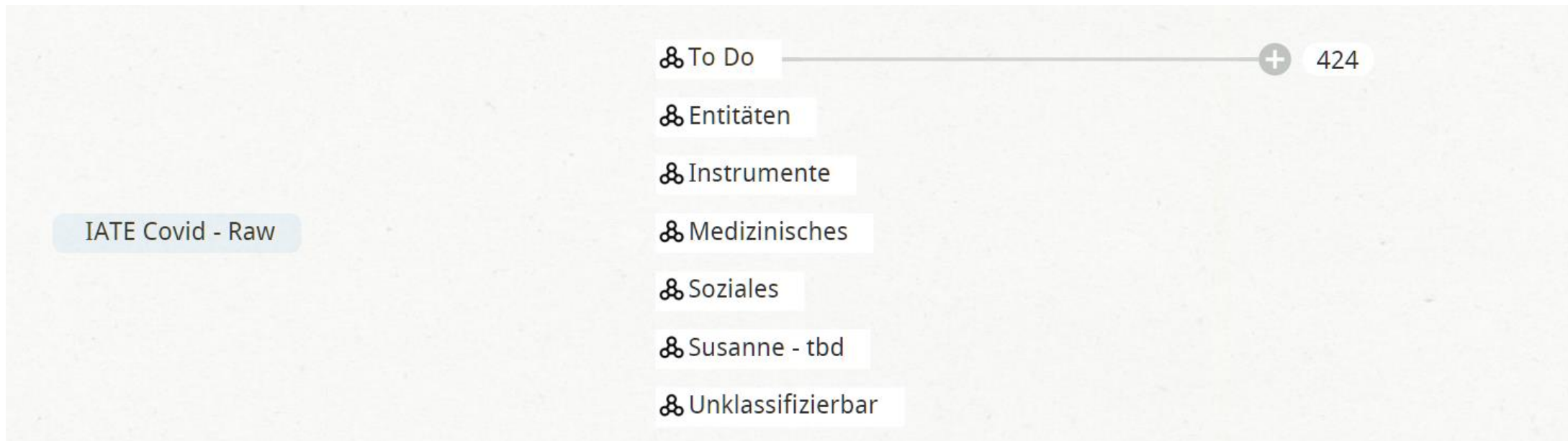
INPUT: a flat set of COVID concepts, no relations between them

OUTPUT: a hierarchical knowledge graph,
consumable via REST/SPARQL

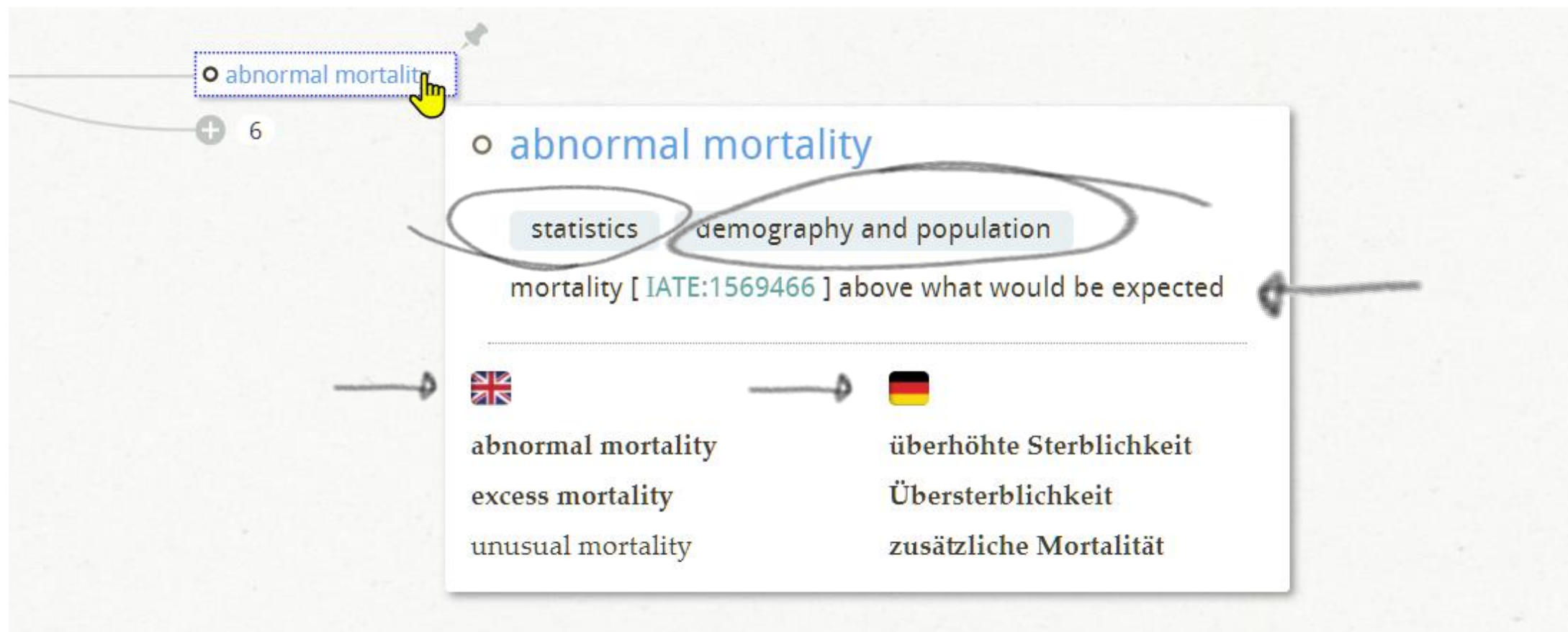
Two Tested Approaches



Manual Taxonomization



- ▶ Top level nodes, temporary helper buckets, and lots to do...



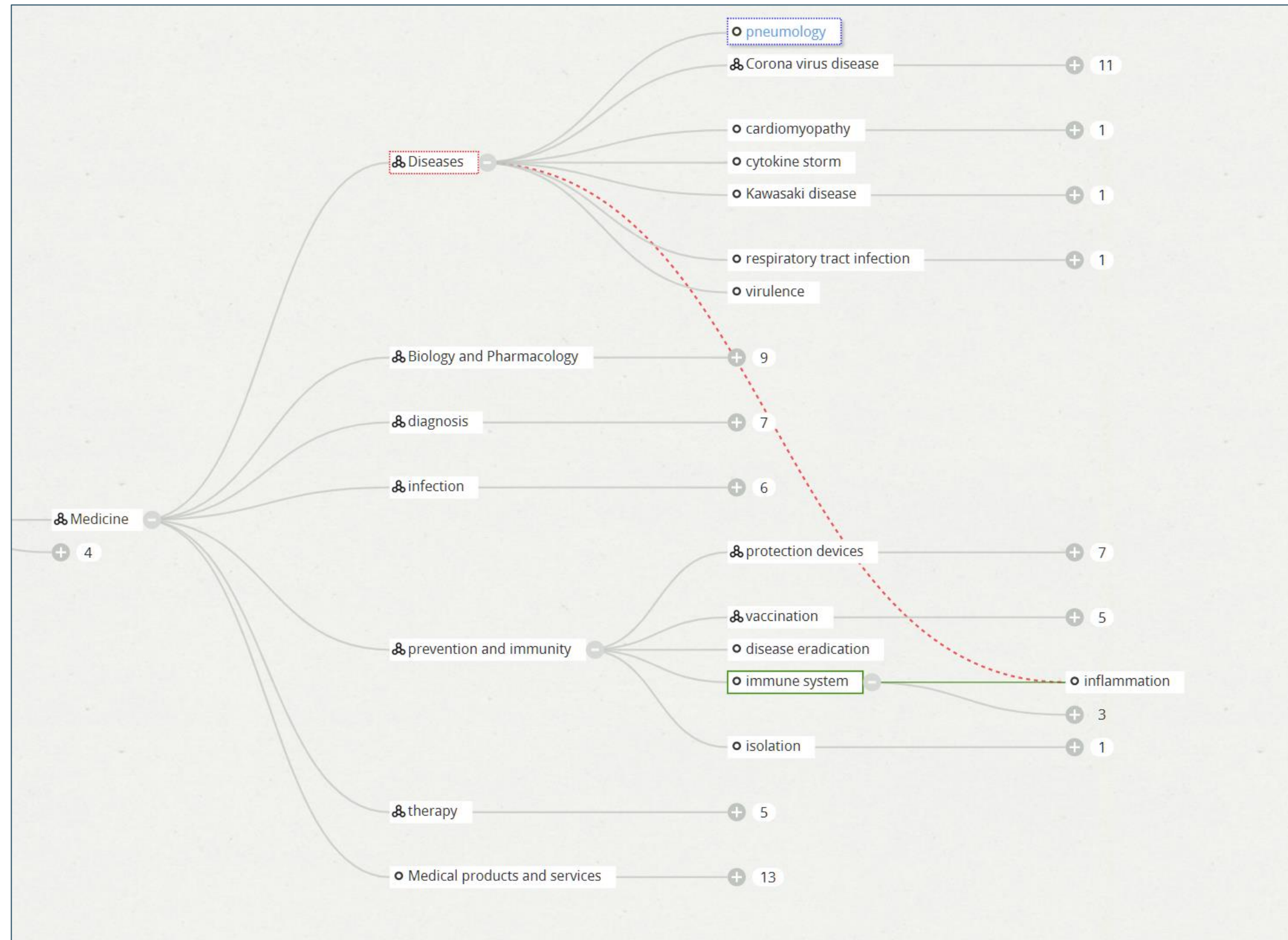
- ▶ Concept card displaying important metadata

Manual Taxonomization: Editing Actions

⌘ drag'n'drop

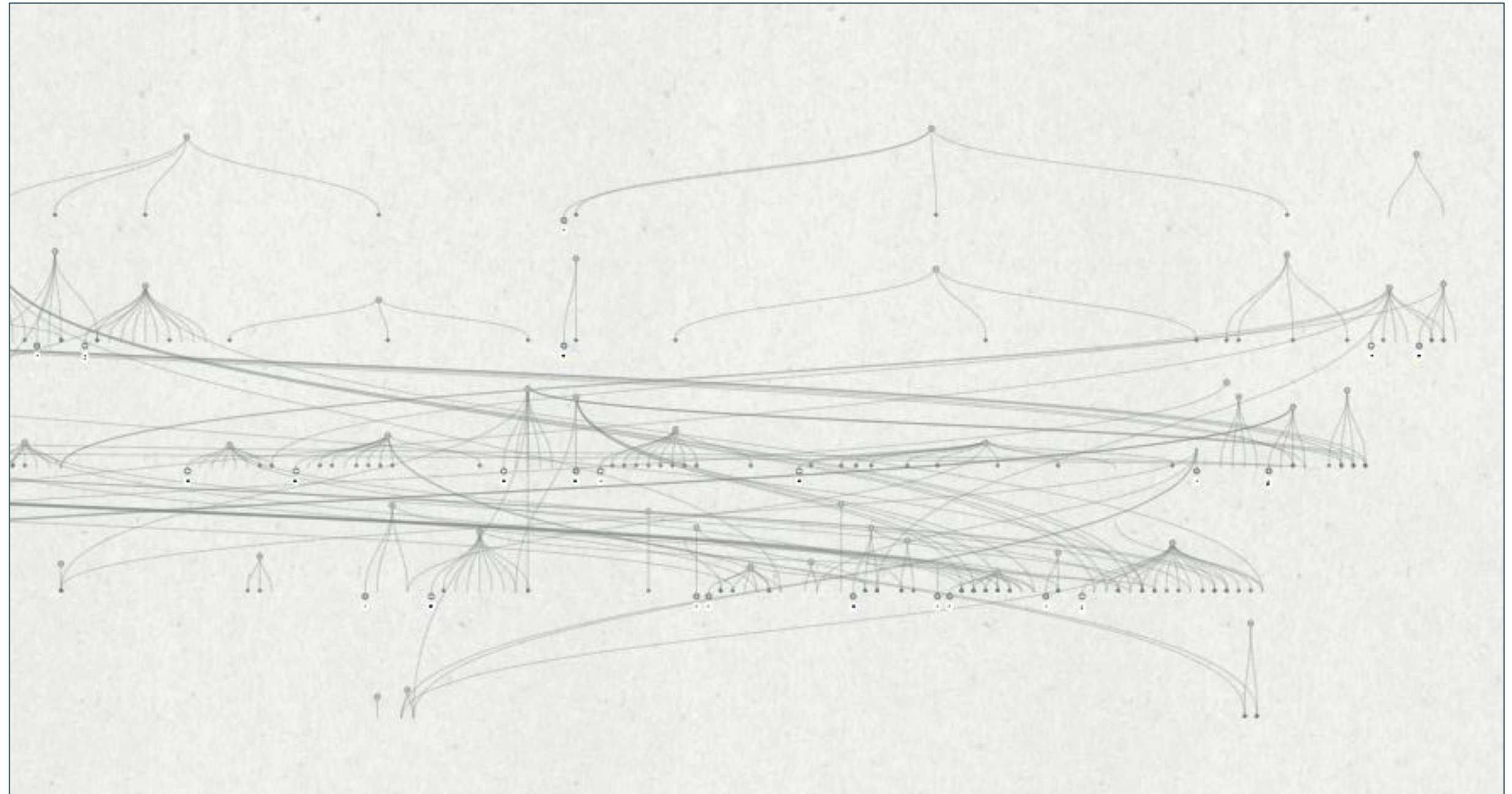
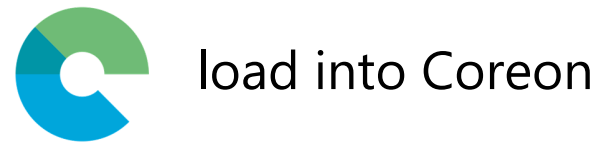
⌘ pin

⌘ filter

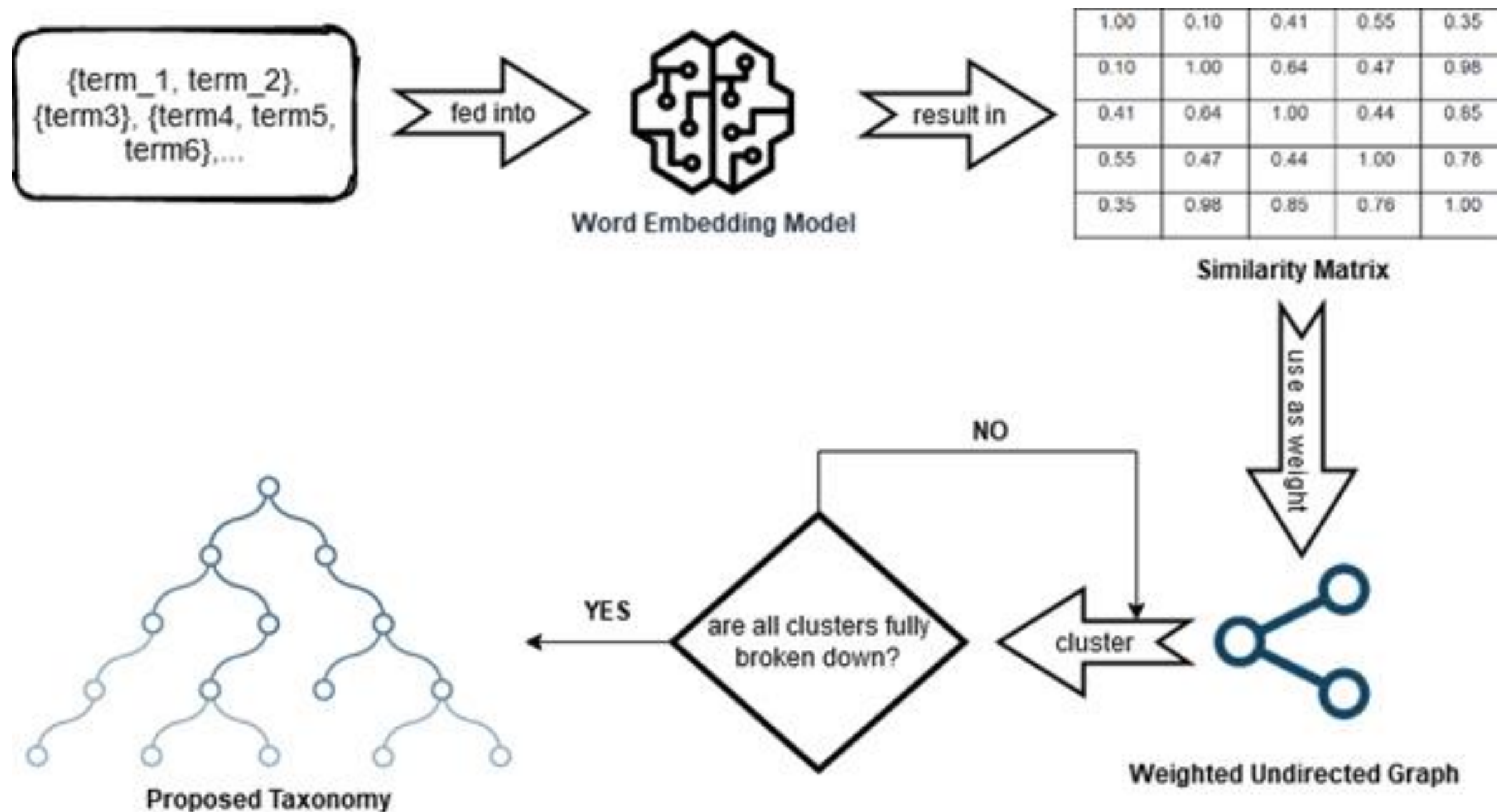


dragging 'inflammation'
from 'diseases'
to 'immune system'

Manual Taxonomization: Result

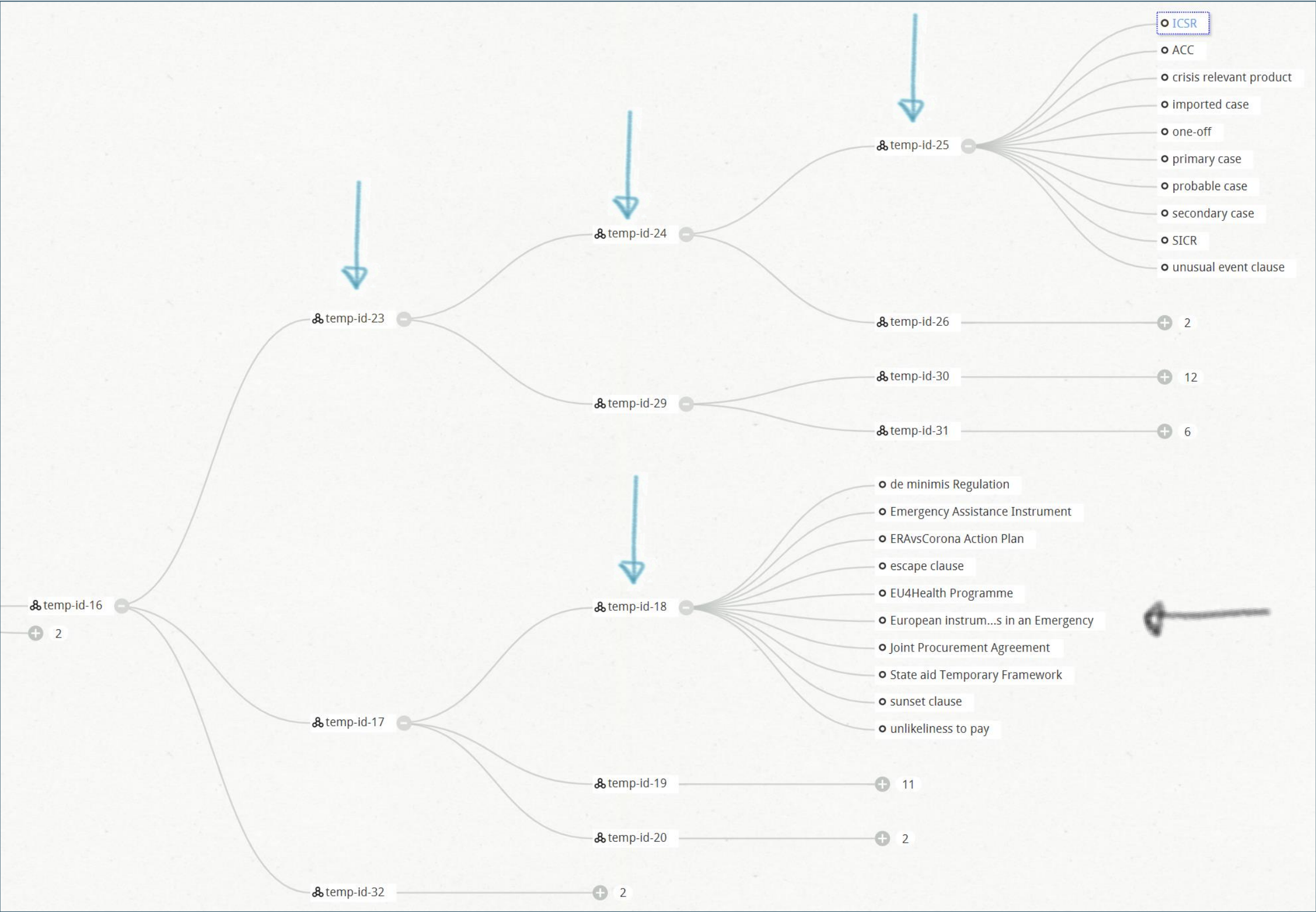


Auto-Taxonomization: Data + fastText WE + Louvain Algorithm



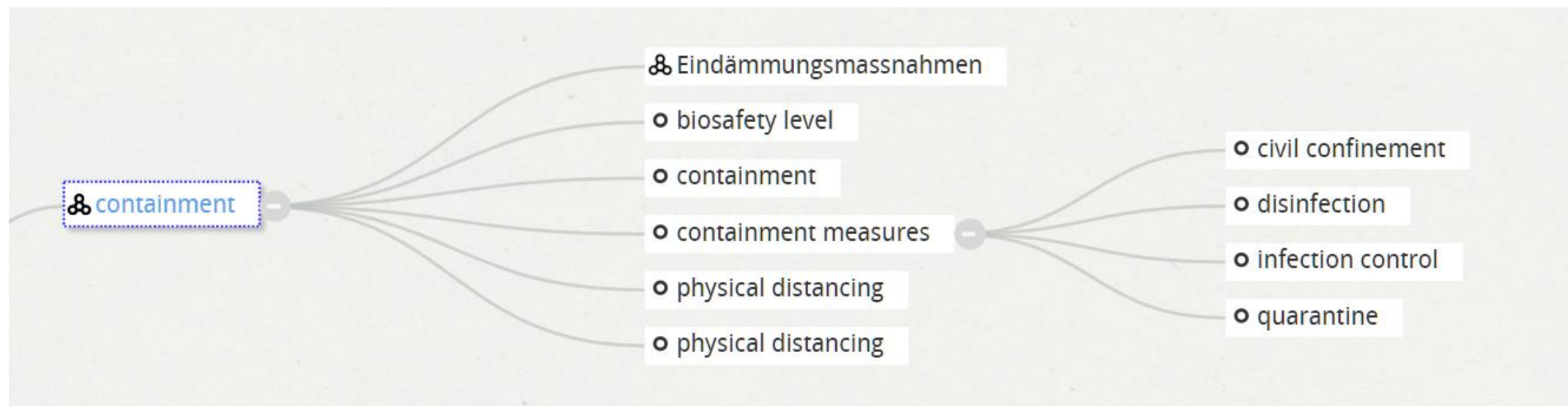
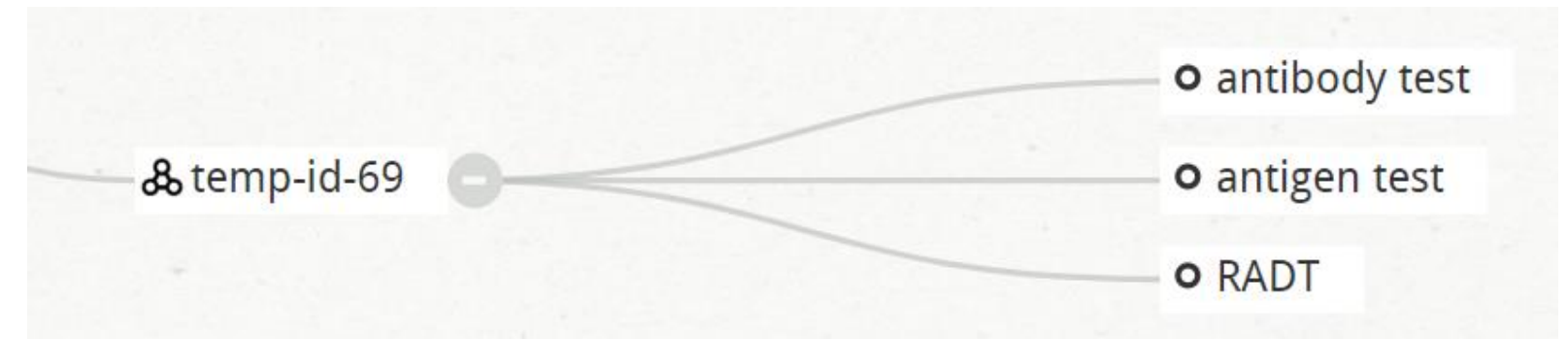
Human Revision of AI-Drafted Taxonomy

initial situation
after automatic
taxonomization



Good Clustering, Bad Clustering?

- ▶ 55 clusters, majority pretty accurate
- ▶ some clusters are off, and we blame **WE**:
 - ▶ ‘interstitial space’ and ‘hospital pharmacy’
 - ▶ spaces appearing in similar “semantic neighborhoods”
- ▶ some existing IATE concepts became parents of concept clusters



Collaborative Taxonomization: Result



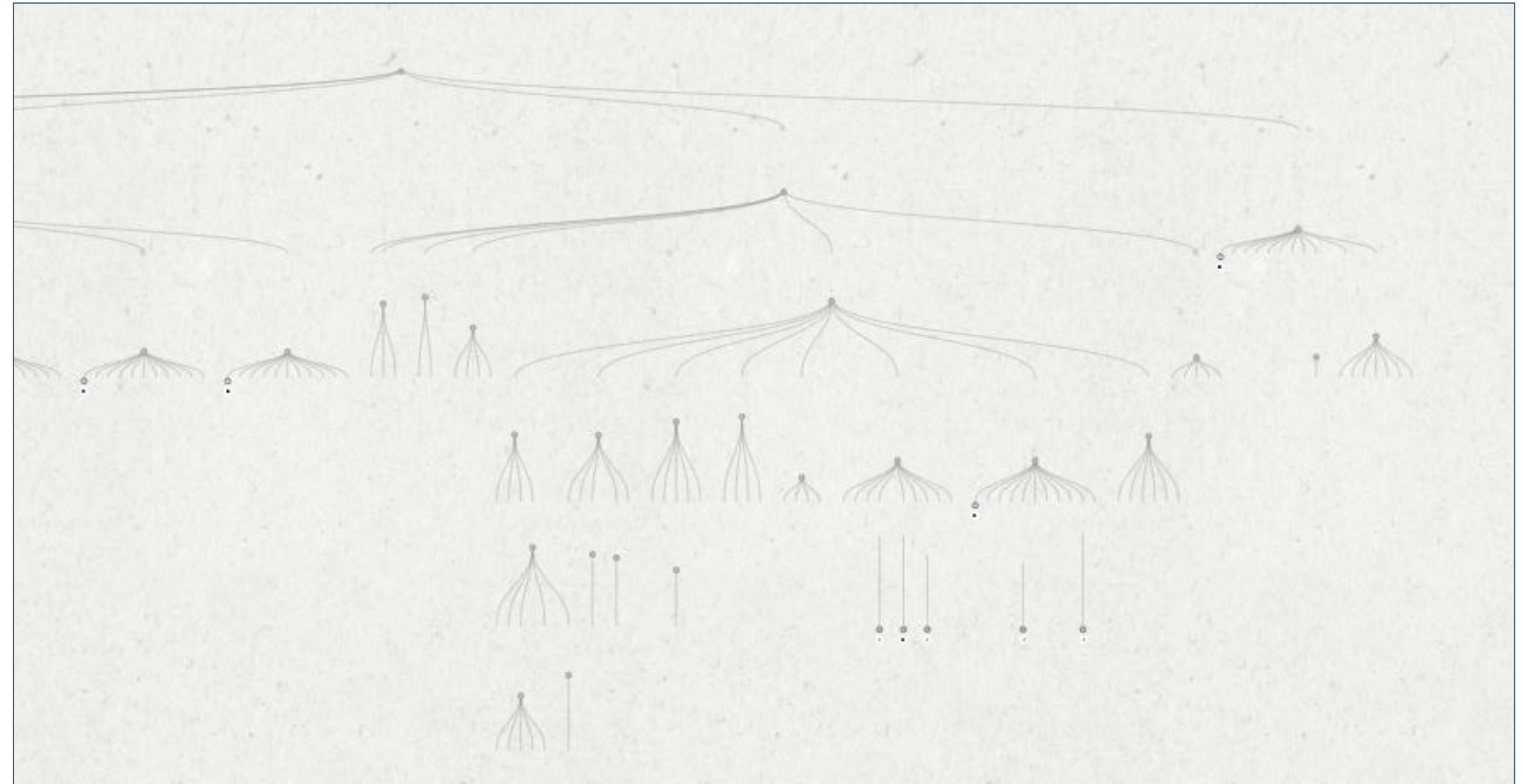
automatic
taxonomization using
ML algorithm



load into Coreon



name auto concepts
move wrong concepts



Metrics	Taxonomization	
	Manual	Semi-Automatic
Curator's recorded time (hours)	40h	8h
Relations created / changed	1 147	432
Concepts created	115	28
Intermediate structural nodes renamed	—	45
Overall relations	679	470

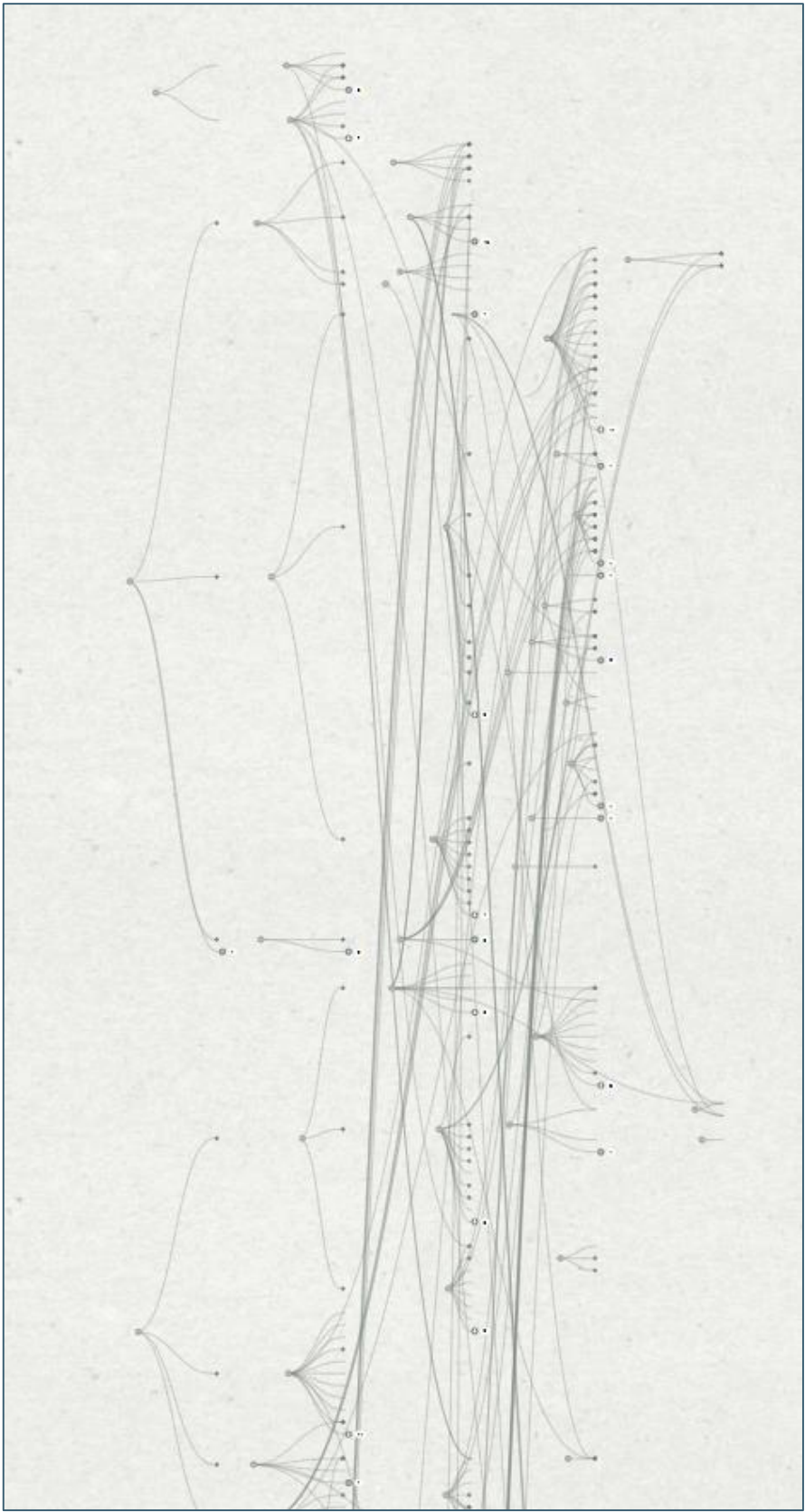
Resulting Taxonomies



load into Coreon



build taxonomy
from scratch



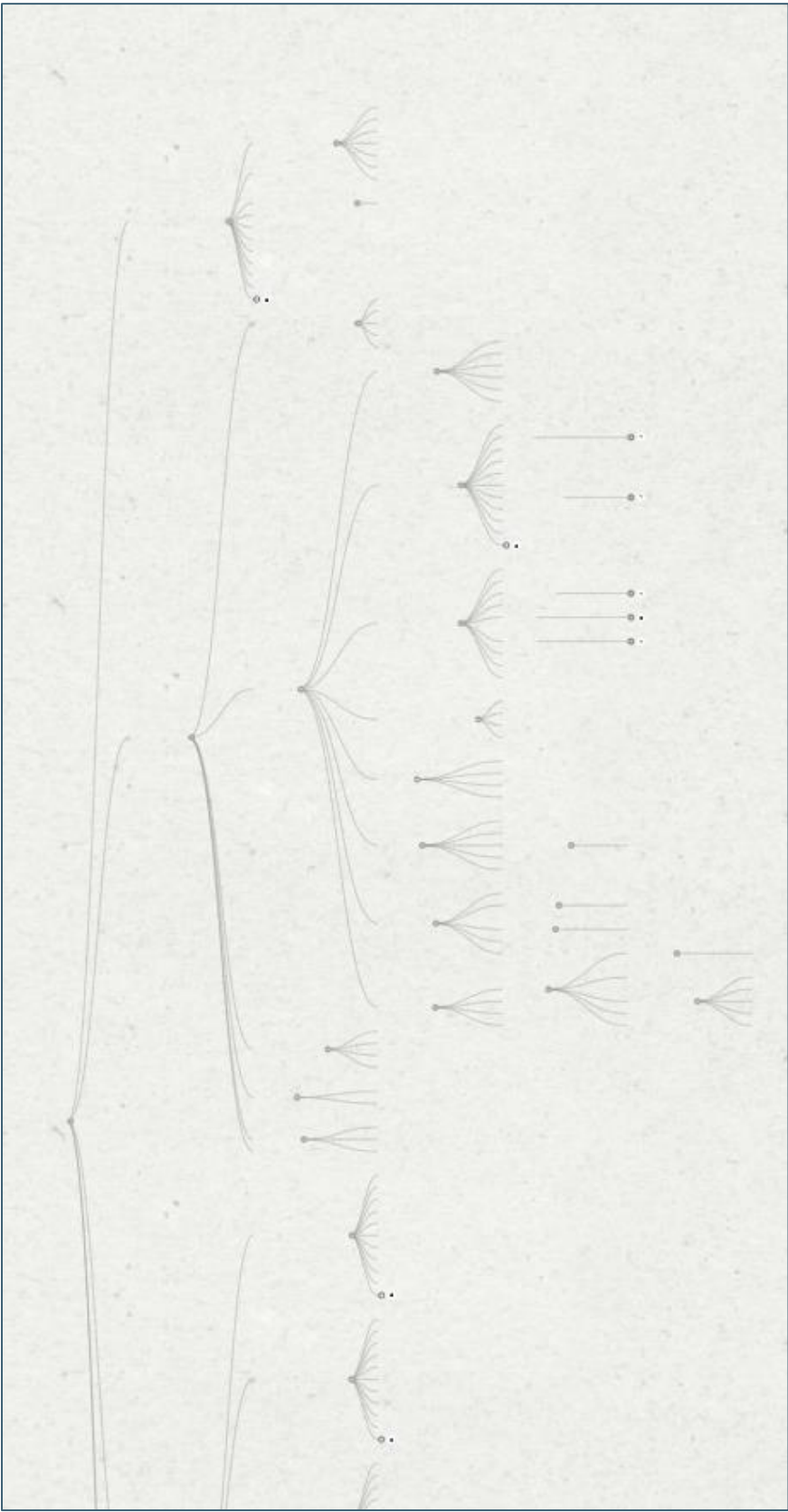
automatic
taxonomization using
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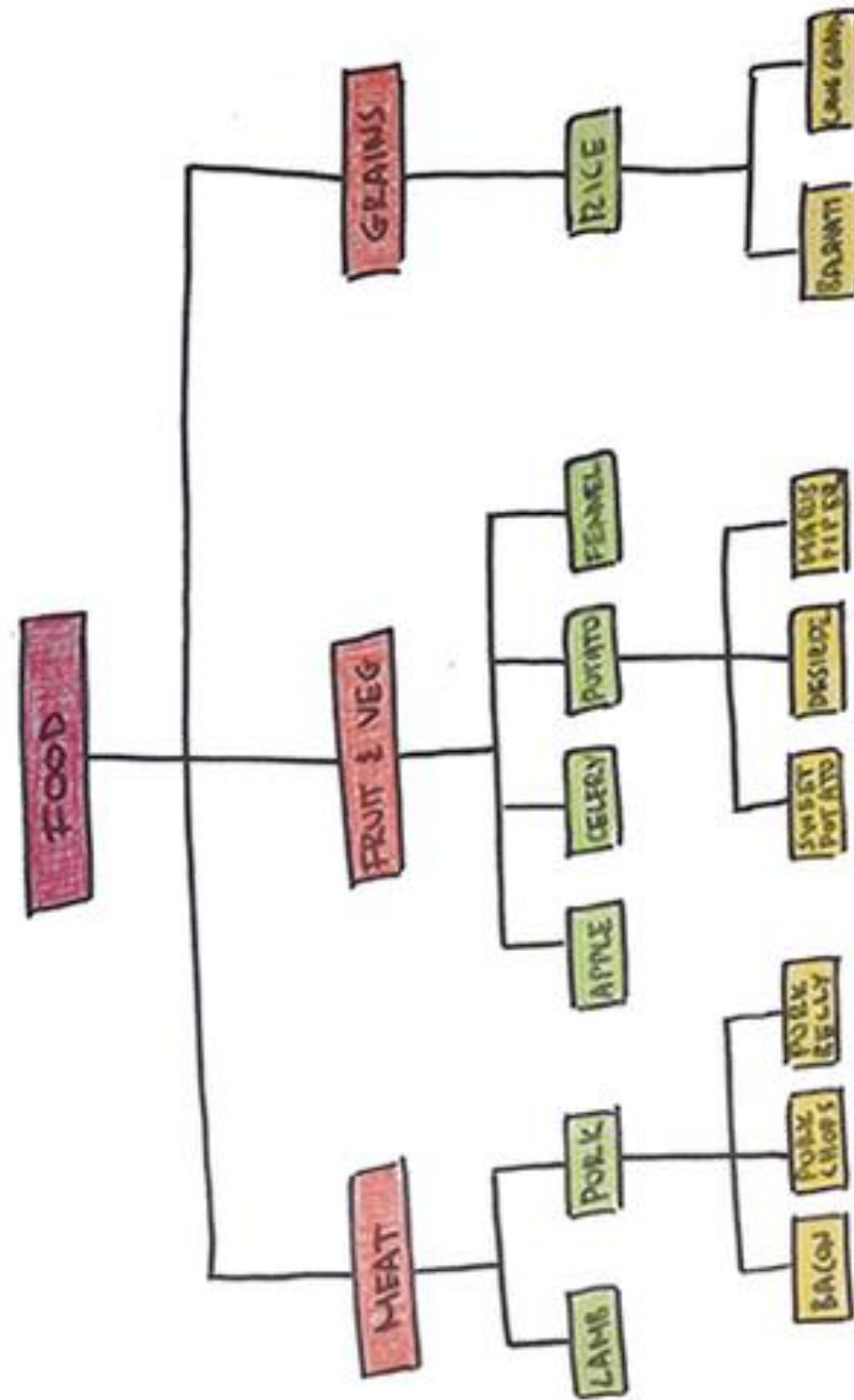
load into Coreon



name auto concepts
move wrong concepts

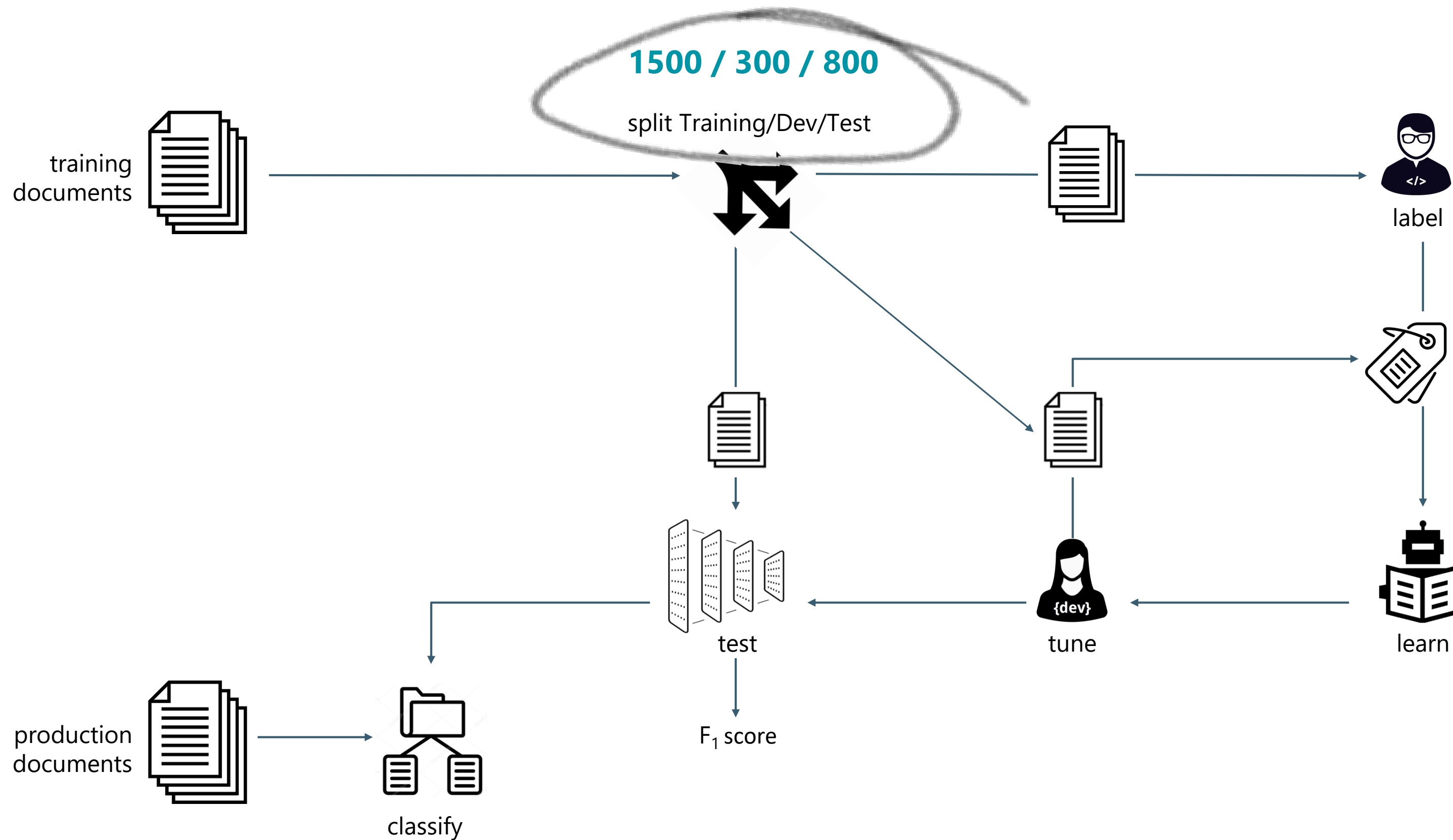


Why Taxonomize?

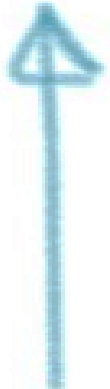


- ⌘ Effective way to add structure to data
- ⌘ Improve data quality
 - ⌘ avoid duplicates and overlapping concepts
 - ⌘ associative relations
- ⌘ Easier and safer data maintenance
- ⌘ Formalize multilingual knowledge, make it machine-digestible
- ⌘ Boost performance of AI algorithms, priming them with structured data

Document Classification



CNN Predictions and True Labels / Test Set

	True labels 	Predicted labels 
The spread of COVID-19 displayed a characteristic sub - exponential linear growth (mean DGR : 0.06 ; range : 0.01 , for Kerala , to 0.11 , for Gujarat , 95 % CI : 0.04 - 0.08 , 11 April 2020 to 3 May 2020) that deviates from exponential growth estimates (mean DGR : 0.1643 ; range : 0.1163 , for Kerala , to 0.2175 , for Tamil Nadu , 95 % CI : 0.1392 - 0.1894 , 2 March 2020 to 3 May 2020) , as a consequence of lockdown strategies () . Reduction of R , from a range of 1.35 - 2.86 (pre - lockdown 95 % CI : 1.49 - 2.21) to 1.13 - 1.67 , in first phase , and to 1.08 - 1.63 , in second phase of lockdowns (post - lockdown 95 % CI : 1.20 - 1.42)...	health management, epidemiology	medicine
"In late 2019 , a cluster of severe respiratory disease cases occurred in Wuhan , China (Hubei Daily (Reporter Yu Jinyi) , 2019 ; World Health Organization , 2020c) . By January 2020 , a novel coronavirus was sequenced , human - to - human transmission was clearly documented (Centers for Disease Control and Prevention , 2020 ; , and additional cases identified in other countries caused by the same virus now known as SARS - CoV-2 , resulting in the identification of a new disease syndrome , . As the virus continued to spread , resources were deployed rapidly to identify and develop prophylactic and therapeutic agents for patients . However , these initial activities were uncoordinated leading to haphazard testing of non - optimal or potentially harmful interventions . One reason for the lack of coordination was the lack of an infrastructure and well vetted guidelines to facilitate optimized drug testing...	medicine, medical products and services	medicine, medical products and services 

Classifiers	Metrics (micro-averages), %		
	Precision	Recall	F-1
Non-initialized CNN	81.6	75.4	78.4
Initialized CNN	82.5	78.8	80.6
📖 Zero-shot 0.95 threshold	15.3	37.8	21.7
📖 Zero-shot 0.97 threshold	15.0	26.3	19.1
📖 Zero-shot 0.99 threshold	12.0	10.2	11.0

Increasing Performance

- Domain-specific data to improve WE (tuning/re-training)
- Leverage metadata
- Exploit concept definitions if available



Modified Live-Virus Vaccine:



en

Definition:

vaccine made from an isolate of an attenuated virus

Definition reference:

COM-EN based on:

Boehringer Ingelheim > Vaccine basics, <http://productionvalues.com/vaccine-basics> [6.1.2010]

Note:

Attenuated means the virus cannot cause disease but it can reproduce in the body cells and stimulate immunity.

Taxonomized Data to Enhance AI Performance



Label documents automatically



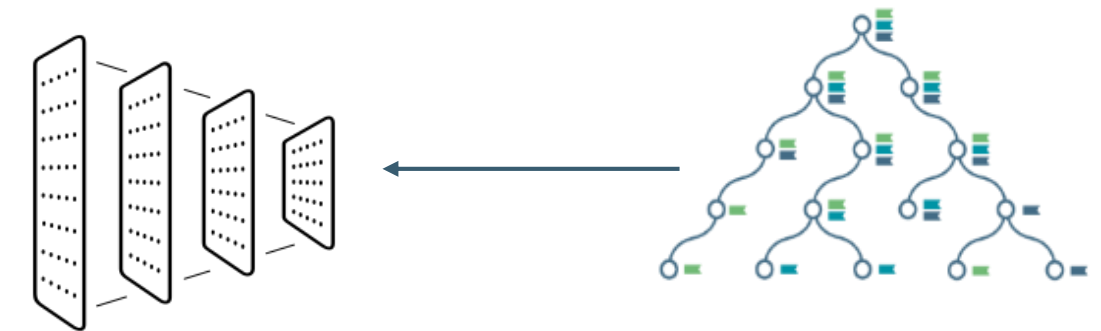
Boost CNN with taxonomy



Enjoy finer granularity in document classification



Get multilingual for free





coreon

Knowledge meets language.

#KGC2021

Thank you!



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